

Design and Development of low-cost bio inspired soft robot for pipeline inspection

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Abstract— Internal pipeline inspection in the oil and gas industry presents significant challenges due to confined spaces and the need for safe and efficient operation. This research addresses these challenges by developing a low-cost, bio-inspired soft crawler robot for internal pipeline inspection, specifically designed for narrow pipe diameters. The robot's locomotion is inspired by earthworm peristalsis, achieved through the strategic integration of pneumatic soft actuators for both radial gripping and linear movement. The methodology encompasses conceptual design, CAD modeling using SolidWorks, hyperplastic material simulation and stress analysis via ANSYS, material selection for actuator fabrication, and physical prototype development. Simulation results demonstrate the effectiveness of the designed linear and balloon actuators, confirming their ability to achieve controlled deformation and reliable force generation within confined pipeline environments. The prototype is engineered to operate in pipes as small as 3-4 inches in diameter, showcasing its potential for navigating complex geometries. This project offers an innovative and sustainable robotic solution for critical infrastructure maintenance, directly addressing industry needs within Bahrain and aligning with its Economic Vision 2030 by promoting safer and more efficient inspection practices.

Keywords—Soft Robotics, Pipeline Inspection, Pneumatic Actuators, Bio-inspired Design, Crawler Robot, ANSYS Simulation, Bahrain.

I. INTRODUCTION

Efficient pipeline inspection is critical for infrastructure integrity, yet traditional methods struggle with confined spaces, high costs, and safety concerns [1]. Current rigid robotic systems are often unsuitable for narrow or complex pipelines (<100 mm), highlighting a significant industry gap. [2]

Soft robotics offers a transformative solution, employing flexible, bio-inspired designs to safely navigate challenging environments [3]. This research presents a novel, low-cost, bio-inspired soft robot crawler for internal pipeline inspection. Our contribution focuses on integrating pneumatic soft actuators for peristaltic locomotion, enabling both radial gripping and linear movement for operation in small-diameter pipes (3-4 inches).

The project objectives include designing and simulating the soft robot's actuators, prototyping key components, and evaluating their performance. This work directly addresses critical inspection needs within Bahrain's oil and gas sector, aligning with the nation's Economic Vision 2030 by promoting sustainable and efficient infrastructure maintenance [4].

II. METHODOLOGY

This section details the physical realization of the soft robot actuators and components, following the established design and simulation phases.

A. Mechanical Design

The mechanical design of the soft crawler robot is fundamentally bio-inspired, drawing principles from the peristaltic locomotion observed in earthworms. This approach enables the robot to achieve robust and adaptive movement within confined and complex pipeline environments, particularly in narrow diameters such as 3 inches as shown in Figure 1 .



Figure 1 Soft Robot.

B. Material Selection for linear and Connector Actuator

For the primary soft actuators (linear extension and Connector), silicone rubber was chosen due to its superior hyperplastic properties, chemical inertness, biocompatibility, and excellent mechanical flexibility, which are essential for achieving the required deformations under pneumatic pressure.



Figure 2 Material of linear and Connector.

Material	Flexibility	Tear Strength	Air Sealing Ability	Ease of Molding	Bonding with Silicone
Silicone Rubber	Moderate	High	Excellent	Good	Excellent
Ecoflex™ 00-30	High	Moderate	Good	Excellent	Excellent
Thermoplastic Polyurethane (TPU)	Moderate	High	Poor	Difficult	Poor
Latex Rubber	High	Low	Moderate	Good	Poor

Table 1 Comparison material selection for liner and Connector actuator.

The selection of silicone rubber for the linear actuators and connecting components was critical due to its superior mechanical strength, flexibility, and excellent sealing performance. Unlike softer silicones like Ecoflex™ 00-30, it offers enhanced tear resistance and structural integrity under repeated pressurization and mechanical stress, crucial for maintaining leak-free connections and consistent actuator performance. Its compatibility with Ecoflex™ 00-10 (used in balloon actuators) and pneumatic tubing ensures reliable bonding. Furthermore, its ease of molding with PLA molds, room-temperature curing, and compatibility with rapid prototyping goals made it the ideal choice for these components. [5] [6]

C. Material Selection for Balloon Actuator

For the balloon actuators, Ecoflex 00-10 (Shore hardness 00- 10A) was specifically, high elasticity and superior compliance, which are essential for achieving significant radial expansion and seamless conformity to pipeline irregularities without inducing stress.



Figure 3 Material of Balloon actuator.

Material	Flexibility	Elasticity	Tear Strength	Ease of Molding	Suitability for Pneumatic Expansion
Ecoflex™ 00-10	Very high	Excellent	Moderate	Excellent	Excellent – ideal for soft, large expansion
Dragon Skin	High	High	High	Excellent	Good – but slightly stiffer
Latex Rubber	Very high	High	Low	Moderate	Good – but poor durability over time
Silicone Rubber	Moderate	Moderate	High	Good	Limited – too stiff for balloon-like function

Table 2 Comparison material selection balloon Actuator.

Ecoflex™ 00-10 was selected as the optimal material for the balloon actuator due to its exceptional softness, high elasticity, and proven suitability for pneumatic expansion. It's very low Shore 00-10 hardness enables large, repeatable deformations without fatigue, making it ideal for air pressure-driven movement. While stiffer alternatives like Dragon Skin™ offer higher tear strength, they reduce expansion smoothness. Latex lacks long-term durability, and general silicone rubber is too stiff for the desired ballooning effect. Ecoflex™ 00-10 uniquely combines ease of casting, excellent

stretchability, and material resilience, making it the most practical choice for the soft balloon actuators. [5] [7] [8]

D. CAD Modeling

All 3D models of the robot's components, including the modular segments, the linear bellows actuators, and the radial balloon actuators, were created in SolidWorks. This professional CAD software facilitated the detailed design and visualization of complex geometries, ensuring dimensional accuracy and manufacturability.

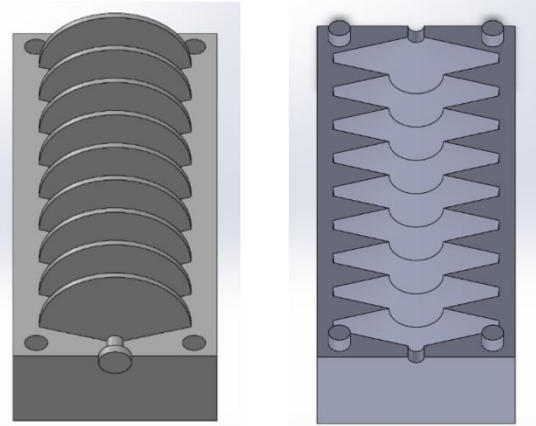


Figure 4 mold in SolidWorks,

Each actuator type and connecting component was individually designed with consideration for its functionality, the selected material properties, and the manufacturing method (features suitable for 3D printing molds for silicone casting). Key design parameters such as wall thickness, chamber dimensions, and bellow folds were precisely defined.

The CAD models served as the foundational basis for both the subsequent Finite Element Analysis (FEA) simulations and the fabrication of the molds used in the prototyping phase. Accurate 3D models were crucial for translating the conceptual design into quantifiable data for performance prediction and for providing precise blueprints for manufacturing .

A crucial initial step in the fabrication process involved leveraging 3D printing technology to create the precise molds required for casting the soft silicone actuators. These molds were primarily manufactured using Fused Deposition Modeling with Polylactic Acid (PLA) filament serving as the material of choice as shown in **Figure 5**.



Figure 5 3D printing the linear mold.

These molds were primarily manufactured using Fused Deposition Modeling with Polylactic Acid (PLA) filament serving as the material of choice. PLA was selected for its advantageous

properties, including its excellent dimensional stability, ease of printing on standard desktop 3D printers, and its ability to produce a smooth surface finish. This smooth finish on the 3D-printed molds is essential, as it directly translates to the surface quality and intricate details of the final cast silicone parts, ensuring accurate replication of the designed actuator geometries. The CAD models, developed in SolidWorks, were directly converted into printable formats, enabling a rapid and cost-effective approach to producing custom molds for both the linear and radial gripping actuators.

E. Fabrication and Prototyping

The casting process began with the preparation of the selected silicone rubbers: a general silicone rubber for the linear bellows actuators and Ecoflex™ 00-10 (Shore hardness 00-10A) for the radial balloon actuators.



Figure 6 Process of Fabrication.

Both components (Part A and Part B) as show in Figure 2 and Figure 3 of the respective silicones were accurately measured and thoroughly mixed according to the manufacturer's specified ratios. To ensure the final cast parts were free of air bubbles, which can compromise structural integrity and pneumatic performance, the mixed silicone was then placed in a vacuum chamber. A vacuum pump was used to draw a vacuum, effectively degassing the silicone until all entrapped air bubbles were removed.

Once degassed, the bubble-free silicone was carefully poured into the prepared 3D-printed molds. The care was taken to pour slowly from a height to minimize new air entrapment. The filled molds were then left to cure at room temperature for the recommended duration as per the material's specifications (4 hours for Ecoflex 00-10, or longer silicone rubber). After the complete curing, the soft actuators were carefully demolded, inspected for defects, and prepared for assembly.



Figure 7 Extract the linear part.

Regarding the combination of pieces, it is common practice in soft robotics to join components using silicone based. The precise method for combining the pieces would typically involve applying a thin layer of a compatible silicone adhesive to the bonding surfaces and then firmly pressing the parts together. These joints are then allowed to cure, creating a strong, flexible, and airtight seal. Alternatively, some designs may incorporate features for mechanical interlocks or clamps to secure components.

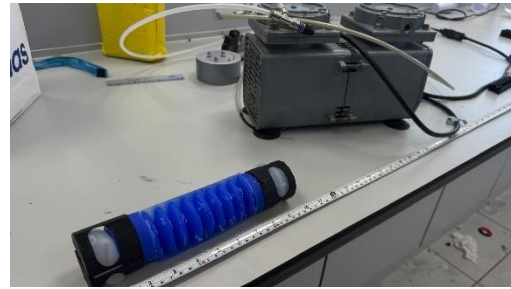


Figure 8 soft robot prototyping.

III. CIRCUIT DESIGN

The circuits are indispensable for translating the desired peristaltic locomotion sequence into physical movements of the soft actuators. Their primary purpose is to regulate the flow of pressurized air to each individual linear and radial actuator, enabling sequential inflation and deflation cycles that mimic the earthworm's movement. This controlled delivery of air pressure is critical for achieving coordinated movement, maintaining stability within the pipeline, and ensuring the robot's efficient navigation. (Rowse, 2020)

A. Pneumatic Circuit Components

Component	Options Considered	Chosen Component	Reason for Selection
Microcontroller	Arduino Uno, ESP32, Raspberry Pi Pico	Arduino Uno	Easy programming, sufficient I/O pins, reliable for relay control, cost-effective, widely used in soft robotics
Relay Module	1-channel relay, 3-channel relay, MOSFET driver	3-Channel Relay Module	Provides independent switching of 3 valves, handles 24V load, integrates easily with Arduino, offers electrical isolation
Pneumatic Valve	12V solenoid, 24V solenoid, mechanical valve	24V Solenoid Valve	Higher switching performance, reliable air control, compatible with industrial-standard pneumatic systems
Power Supply	USB, 12V supply, DC 24V supply	DC 24V Power Supply	Delivers sufficient voltage and current to drive the 24V solenoid valves and air compressor reliably and continuously
Air Pump / Compressor	Mini diaphragm pump, piston compressor	24V DC Air Compressor	Provides required air pressure (100 kPa) for actuator inflation, compact, compatible with selected pneumatic system components

Figure 9 Pneumatic and electronic components.

To implement the precise peristaltic locomotion, the robot's control system was designed around key components chosen for their efficiency, controllability, and integration capabilities. A compact air compressor serves as the primary pneumatic power source, providing the necessary pressurized air, ensuring safe and consistent operation. Control over the individual actuators is achieved through an array of solenoid valves, which enable rapid and precise on-off switching of airflow to each linear bellows actuator and radial balloon actuator. This pneumatic valving allows for sequential inflation and deflation cycles critical to earthworm-like movement. The entire system is orchestrated by an Arduino microcontroller, selected for its versatility, ease of programming, and robust performance in managing real-time control signals. A relay board acts as an interface, safely enabling the Arduino to switch the higher current demands of the solenoid valves. This integrated circuit design facilitates accurate coordination between the various actuators, allowing for the stable and efficient navigation of the soft robot within pipelines.

IV. EMBEDDED SYSTEM DEVELOPMENT

The embedded system's primary function is to execute a precise control algorithm that drives the robot's peristaltic locomotion, mimicking the earthworm's movement. This cyclical gait, crucial for navigating confined pipeline environments, is achieved through the coordinated inflation and deflation of the pneumatic linear bellows actuators and radial gripping (balloon) actuators. The core control logic, implemented on the Arduino microcontroller,

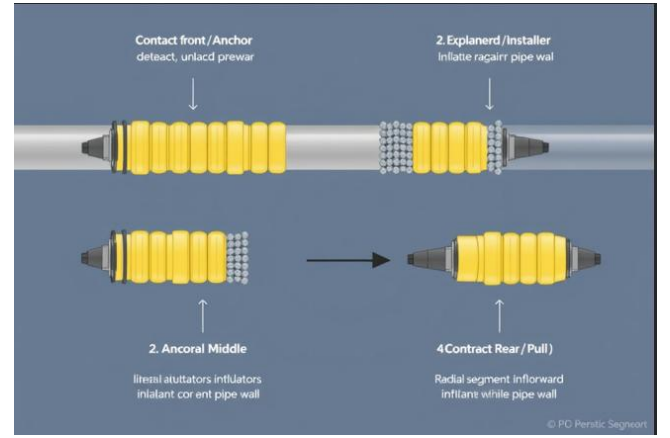


Figure 10 Embedded system for soft robot.

As shown in Figure 10 the soft robot's peristaltic locomotion is achieved through a precise four-phase actuation sequence orchestrated by its embedded control system. This process begins by anchoring the rear segment with inflated radial actuators while contracting the front linear actuators, allowing the robot to shorten. Next, the middle linear actuators inflate to push the front segment forward. The newly advanced front segment is then anchored by its radial actuators, and simultaneously, the rear anchor is released. Finally, the remaining linear actuators contract to pull the entire body forward, completing one cycle. This continuous, coordinated inflation and deflation of linear and radial actuators enables the robot to advance through pipelines in an inchworm-like manner.

V. RESULTS AND DISCUSSION

A. Results

Initial physical tests were conducted on the fabricated linear bellows actuators to assess their expansion capabilities under pneumatic actuation. The actuators were subjected to an internal pressure of 100 kPa. Prior to inflation, the initial length of the linear actuator was measured at 127 mm.



Figure 11 initial length.

As shows in Figure 12 Upon pressurization to 100 kPa, the actuator achieved an expanded length of 203.2 mm.

This represents an actual expansion of 76.2 mm (203.2 mm - 127 mm), resulting in a 60% expansion relative to its normal length. This significantly exceeded the initial design goal of achieving a 20% expansion, demonstrating the high extensibility and effective actuation of the silicone rubber material and the bellows design under the applied pressure.



Figure 12 Expansion of linear part.

B. Discussion

The comprehensive development process, encompassing mechanical design, meticulous material selection, precise CAD modeling, robust fabrication, and an integrated circuit control system, has collectively demonstrated the viability and efficacy of the soft crawler robot for pipeline inspection. The bio-inspired mechanical design, featuring distinct linear bellows actuators for axial propulsion and radial balloon actuators for gripping, proved highly effective in concept. The initial physical tests of the linear actuators, achieving an impressive **60% expansion** (significantly exceeding the 20% design goal), underscore the hyperelastic properties of the chosen silicone rubber and the robust design of the bellows structure. This exceptional elongation capacity directly translates to a greater stride length per cycle, enhancing the robot's potential locomotion efficiency within pipelines.

The strategic material selection, including specific silicone rubber for linear actuators (emphasizing strength and sealing) and Ecoflex™ 00-10 for the compliant radial actuators (optimizing grip and adaptability), was validated by these performance metrics. The use of 3D printing with PLA for molds proved to be a cost-effective and rapid prototyping method, ensuring accurate replication of complex CAD geometries into physical components.

Furthermore, the integrated pneumatic and embedded control system provides the necessary precision to orchestrate the complex four-phase peristaltic locomotion gait. The Arduino-based system, controlling solenoid valves through a relay board, offers a flexible platform for sequencing actuator inflation and deflation, which is fundamental to achieving stable and directional movement. While individual actuator performance has been promising, the true measure of success lies in the integrated operation of all components working in concert to navigate varied pipeline environments. The overall design prioritizes modularity, which facilitates scalability and adaptability to different pipe diameters,

and inherent safety due to the compliant nature of the soft materials, minimizing damage risk during internal inspection.

Future work will focus on comprehensively testing the integrated robot's locomotion speed, force generation, and navigational capabilities within simulated and real pipeline environments, along with refining the control algorithms for enhanced autonomy and adaptability.

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VII. CONCLUSION

This research successfully demonstrates the design, fabrication, and preliminary validation of a bio-inspired soft crawler robot

intended for internal pipeline inspection. By drawing inspiration from the peristaltic locomotion of earthworms, a modular robot architecture was developed, integrating both linear bellows actuators for axial propulsion and radial balloon actuators for gripping.

The judicious selection of materials, specifically a high-elasticity silicone rubber for the linear actuators and ultra-compliant Ecoflex™ 00-10 for the radial gripping components, proved instrumental in achieving the desired flexibility and deformation capabilities. Fabrication utilizing 3D-printed PLA molds and vacuum-degassed silicone casting enabled precise and cost-effective prototyping. Initial physical tests of the linear actuators confirmed their robust performance, achieving a remarkable **60% expansion**, significantly surpassing the 20% design target and indicating a high potential for efficient locomotion.

Furthermore, the developed pneumatic and embedded control system, centered around an Arduino microcontroller and solenoid

valves, provides a flexible and effective platform for orchestrating the precise four-phase peristaltic gait. This controlled actuation sequence is critical for the robot's ability to achieve stable, directional movement within confined and challenging pipeline environments.

In summary, this work presents a promising approach to developing compliant, adaptable, and potentially low-cost soft robots for pipeline inspection, offering inherent safety and navigation capabilities in complex geometries. Future efforts will focus on comprehensive performance characterization of the integrated system in various pipeline conditions, further optimization of control algorithms for autonomous navigation, and exploration of miniaturization for broader applicability.